

SOARING INTO SPACE

ROCKETS

In order to escape Earth's atmosphere and enter orbit, a space rocket must reach speeds greater than 17,000 mph (25,000 km/h). The enormous force needed to accelerate a rocket to such speeds is provided by extremely powerful engines. These generate thrust by producing huge amounts of hot gas in extremely rapid chemical reactions. The thrust overcomes the force of the rocket's weight so that it can lift off into space.



SPEED DEMON

British car *Bloodhound* can accelerate to more than 630 mph (1,000 km/h) thanks to its huge jet engine. It is hoped it will one day break the land speed record. Jet engines are similar to rocket engines but burn fuel using oxygen from the air.

Blastoff

A Proton-M rocket lifts off from the Baikonur Cosmodrome in Russia. Extremely rapid chemical reactions between its fuel and another substance, called an oxidizer, produce expanding exhaust gases that push out through its nozzles.

UNBALANCED FORCES

Before a rocket launches, the forces acting on it are balanced—the upward force (produced by the launchpad) is equal to the downward force (the weight of the rocket). For it to take off, the rocket's thrust must be greater than the rocket's weight.

